ME 165
Basic Mechanical Engineering

Lecture 11

Md. Aminul Islam
Lecturer
Department of Mechanical Engineering, BUET

Refrigerants

	R22	R134A	R410A
Boiling Point:	-40.8 °C -41.40 °F	-26.3 °C -15.34 °F	-48.5 °C -55.4 °F

Boiling point of Water: 100 °C
212 °F

Refrigerants

Desirable Properties of Refrigerants:

The important properties of refrigerants that relate to the overall performance of a refrigeration system are given below –

- A high latent heat of vaporization is desirable because it is usually associated with a high refrigerating effect per unit mass of refrigerant circulated.
- A high critical temperature is desirable as it is impossible to condense the refrigerant at a temperature above the critical, no matter how much the pressure is increased.
- A refrigerant should have a low boiling temperature; Otherwise it would become necessary to operate the compressor at high vacuums with the resulting lowered efficiency and capacity.
- The freezing point of the refrigerant should be lower than the lowest operating temperature of the cycle to prevent blockage of refrigerant pipelines.
- Low refrigerant densities are usually preferable, since they permit the use of small suction and discharge lines without excessive pressure drops.

Refrigerants

Desirable Properties of Refrigerants:

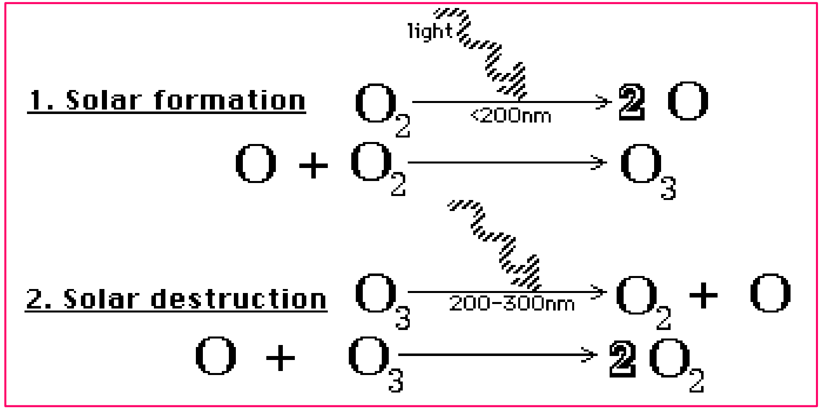
- A refrigerant should have low volume per unit mass when in gaseous state. This not only reduces the size of the equipment but also means higher compressor efficiency
- The refrigerant should have a lower viscosity and high values of thermal conductivity.
- The refrigerant should be chemically stable in operating conditions
- The refrigerant should have no chemical reaction with the lubricating oil.
- The refrigerant should be non toxic, non corrosive and inert.

Ozone Layer Depletion

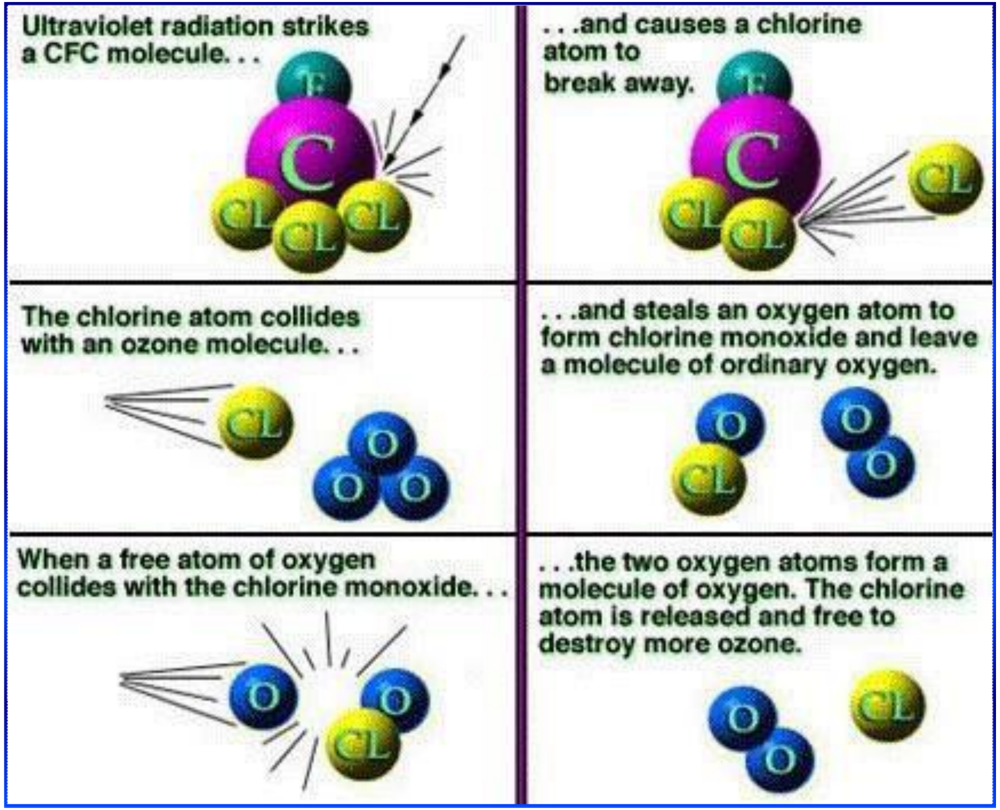
- The ozone layer is found in the lower portion of the stratosphere, approximately 15-35 km above earth surface.
- The ozone layer contains about 10 ppm of ozone, while the average ozone concentration in atmosphere as a whole is about 0.3 ppm of air.
- The ozone layer in the stratosphere block many of the sun's harmful rays (UV-A and UV-B).
- The thicker the layer of stratospheric O_3, the greater the protection.

Formation and Destruction:

Being a natural constituent of the stratosphere, O_3 is regularly formed and destroyed in a cyclic manner having a **dynamic steady state.**



Ozone Layer Depletion



Ozone Layer Depletion

CFC's and HCFC's deplete the O_3 layer.

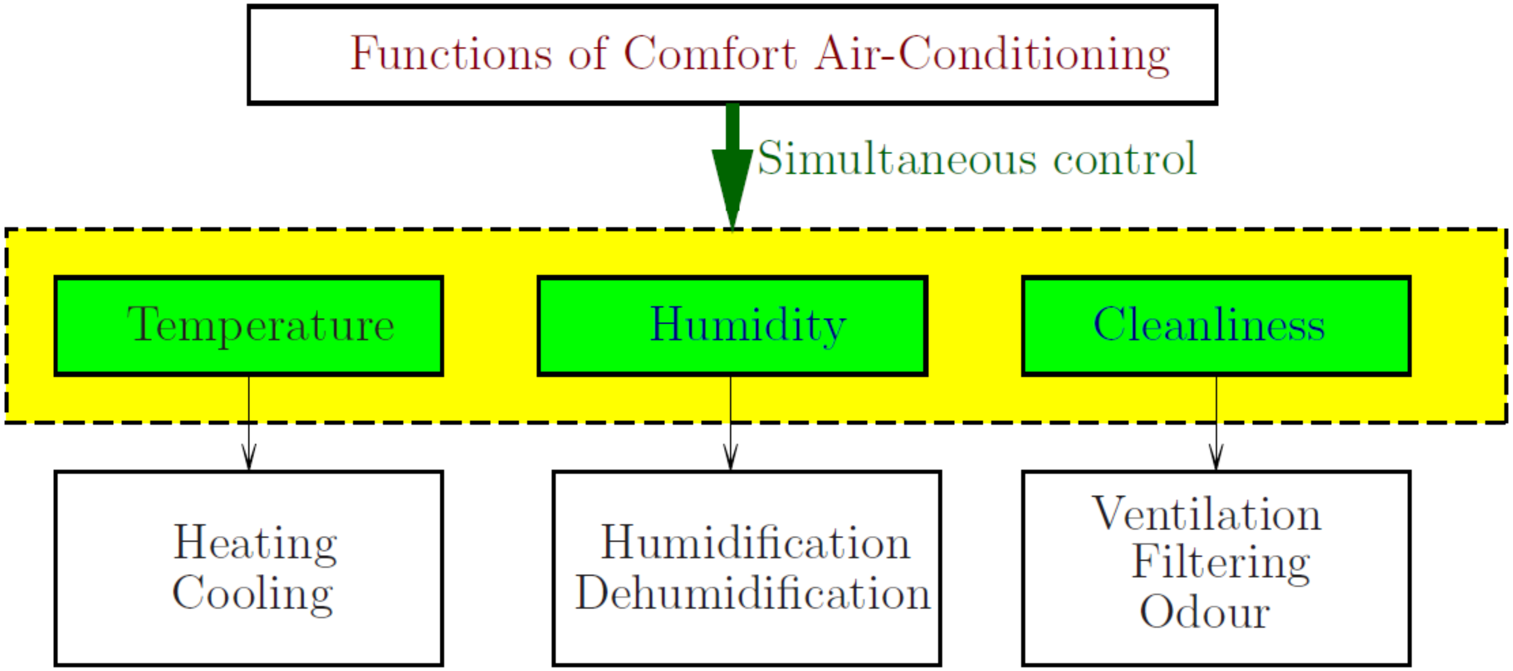
- **They do not occur naturally and are totally man-made,** and are found in air conditioning and refrigeration systems, in industrial manufacturing processes, and in some cleaning solvents.
- The depleted O_3 layer allows too much ultraviolet rays down to the earth's surface, increasing the risk of skin cancer in humans and damaging the environment, especially harming sea life.
- O_3 depletion also leads to a reduction in cropyield.
- CFCs are no longer manufactured in industrialized countries, but remain in older refrigeration equipment including *automobile air conditioners*.
- Available illegally or in the black-markets.

Air Conditioning Systems

Air Conditioning Systems

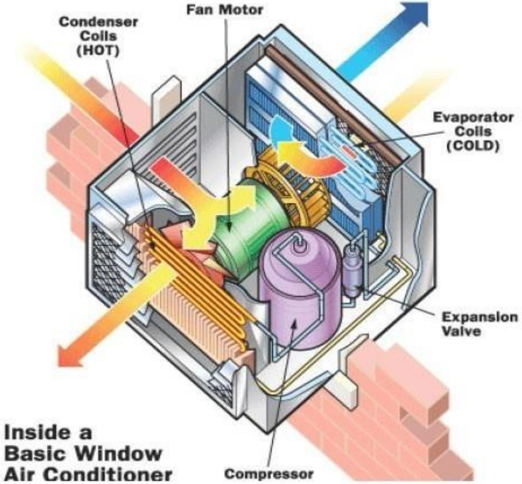
- Air conditioning is the cooling and dehumidification of indoor air for thermal comfort. In a broader sense, the term can refer to any form of cooling, heating, ventilation, or disinfection that modifies the condition of air.
- An air conditioner (often referred to as AC) is an appliance, system, or mechanism designed to stabilize the air temperature and humidity within an area.
- HVAC = Heating, Ventilation and Air Conditioning
- Typical Air Conditioning Systems
  - Unitary System
    - Single Package Unit (Window Type AC)
    - Split System
  - All-Air System
  - All-water System
  - Air-Water System

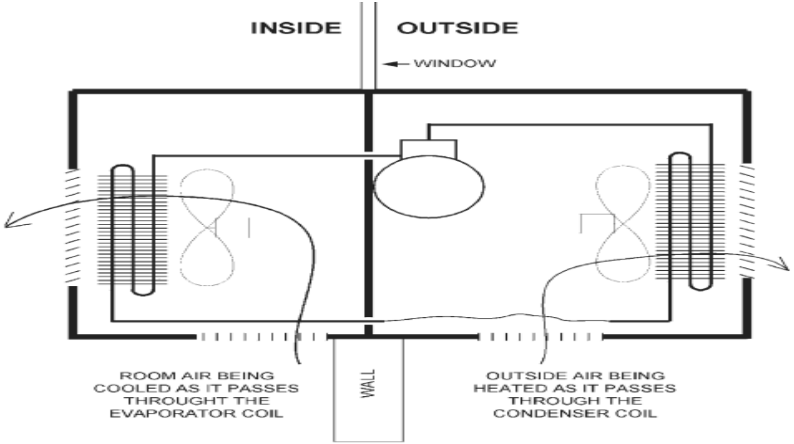
Air Conditioning Systems



Air Conditioning Systems

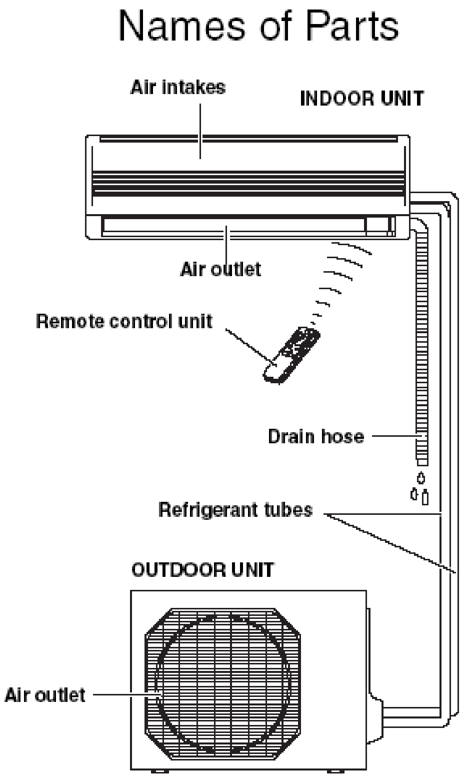
Single Package Unit/ Window Type AC:

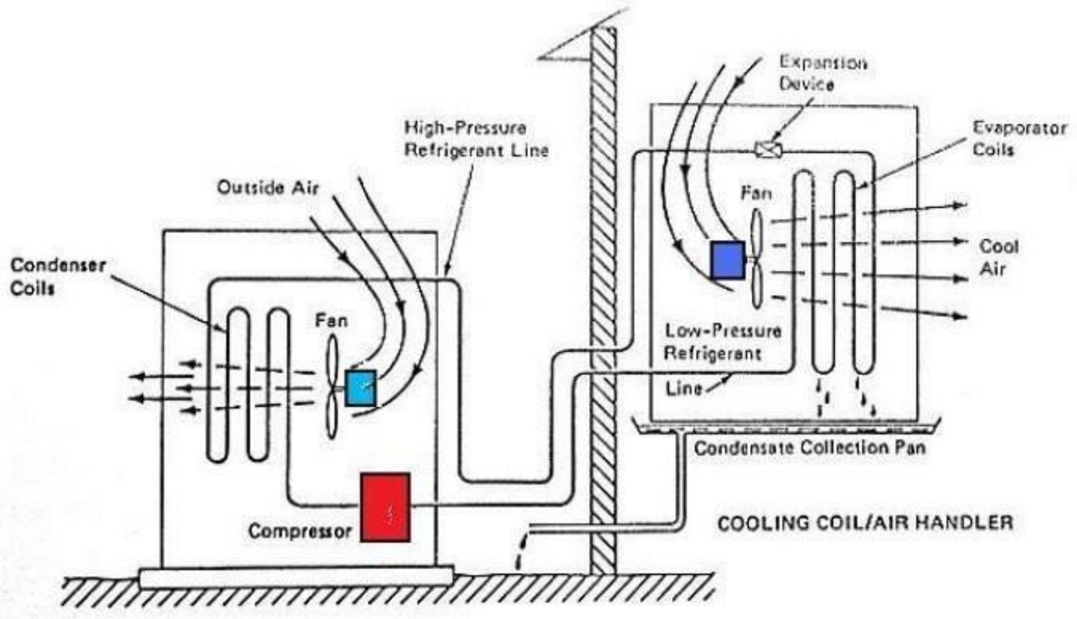




Air Conditioning Systems

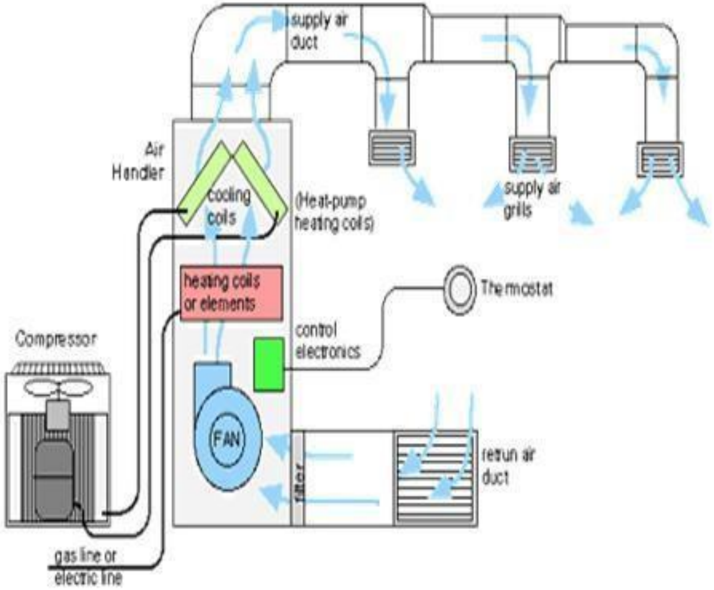
Split Type AC:





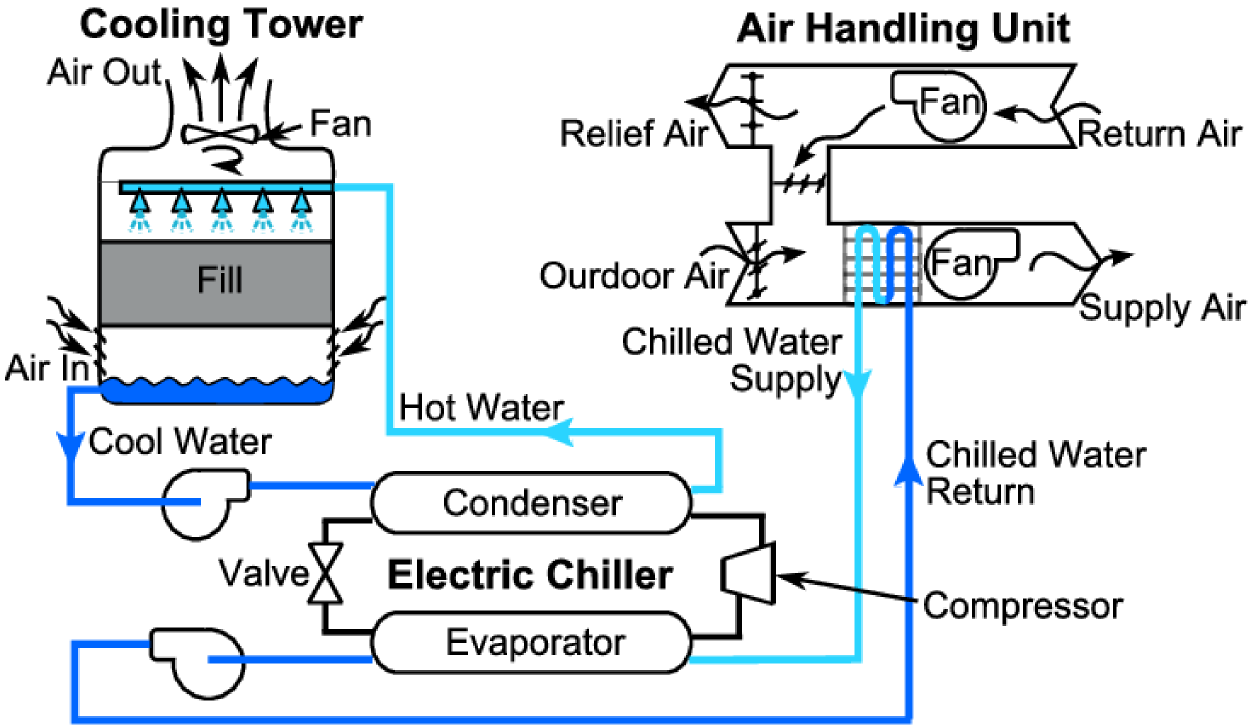
Air Conditioning Systems

Ducted Split Type AC:



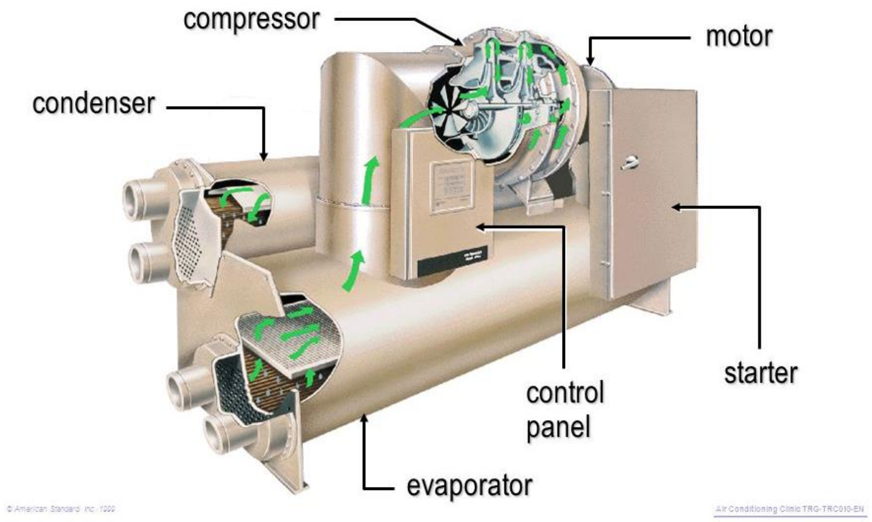
Air Conditioning Systems

Conventional HVAC System (All Air System):



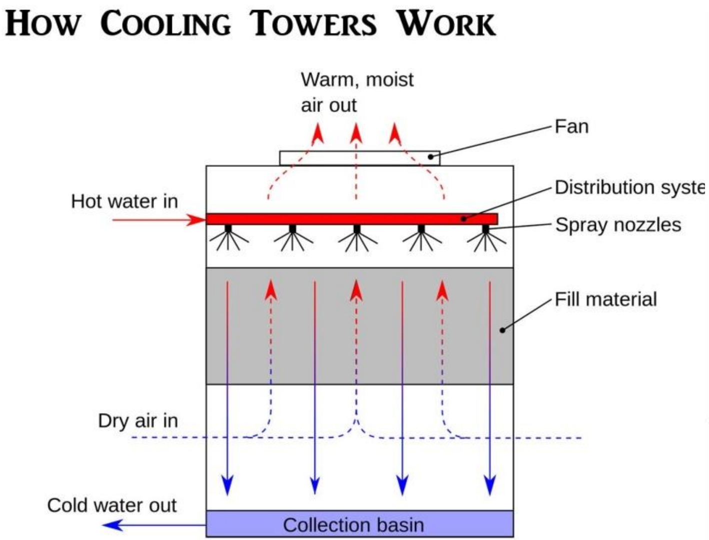
Air Conditioning Systems

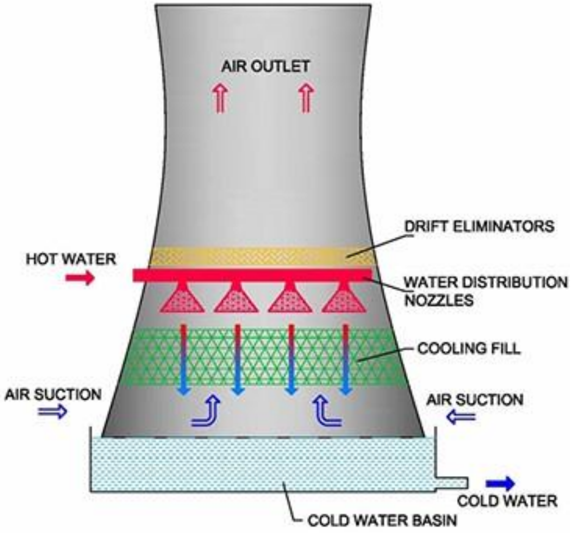
Chiller:



Air Conditioning Systems

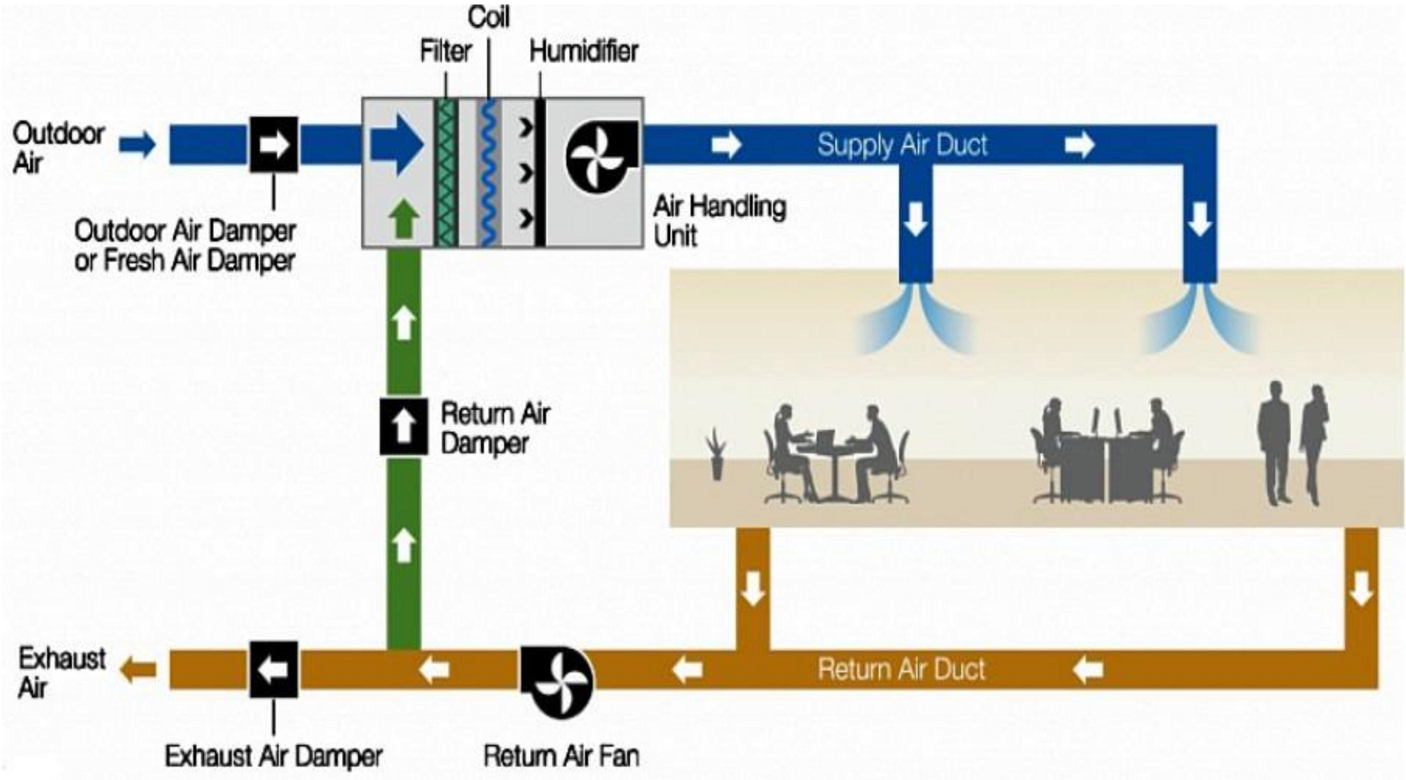
Cooling Tower:





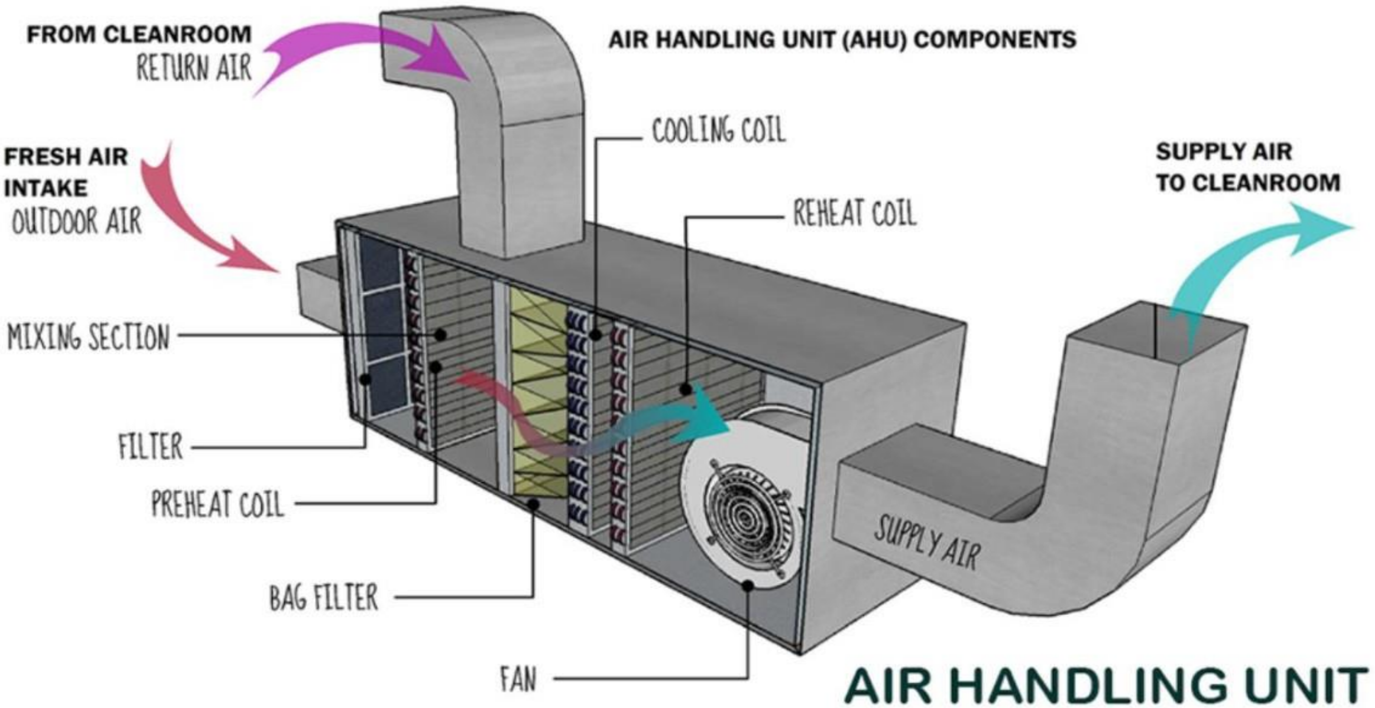
Air Conditioning Systems

Air handling Unit



Air Conditioning Systems

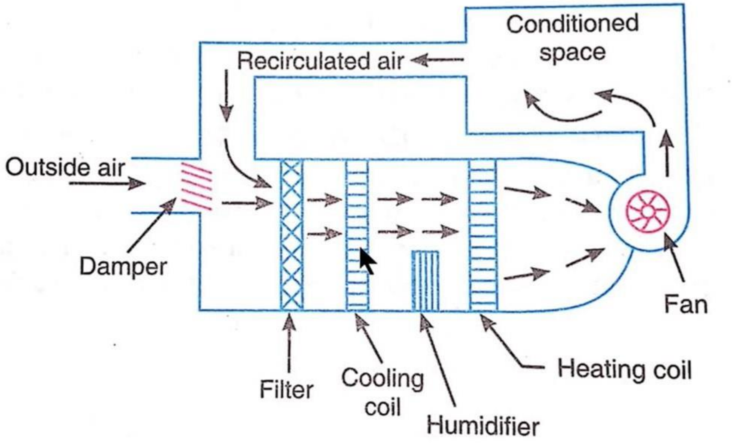
Air handling Unit



Air Conditioning Systems

Air handling unit:

- The AHU is a large metal box containing separate ventilators for supply and exhaust,
  - ✓ heating coil,
  - ✓ cooling coil,
  - ✓ heating/cooling recovery system,
  - ✓ air filter racks or chambers,
  - ✓ sound attenuators,
  - ✓ mixing chamber, and
  - ✓ dampers
- AHUs connect to ductwork that distributes the conditioned air through the building, and returns it to the AHU.
- Depending on the required temperature of the re-conditioned air, the fresh air is either heated by a recovery unit or heating coil, or cooled by a cooling coil.



- The basic function of the AHU is
  - ✓ take in outside air,
  - ✓ re-condition it and
  - ✓ supply it as fresh air to a building.

Thank you